Constraints and source mapping
Train implements Comparable<Train>
Load implements Comparable<Load>
Train: {sum(load.weight) <= maximumLoad}
Train: {dockedAt XOR onTrack}
routeStations: {ordered, unique}; cyclic route >= 3 stations
Load: {waitingAtStation XOR carriedByTrain} after each operation
Track: {at most one track per unordered station pair}
ID-based associations describe business roles, not Java object fields.

Derived docking and notation
Train, TrainStation and Load are abstract (italic class names).
dockedAt is derived from Train.stationId (null while on track).
Each train is docked at 0..1 station; each station has 0..* docked trains.
{numberOfDockedTrains <= maxTrains}
Capacity is checked on arrival; createTrain() does not enforce it.
{readOnly} denotes a final field; final Lists may still be mutated.

TrainsController

- stations : List<TrainStation> = new ArrayList<>()
- tracks : List<Track> = new ArrayList<>()
- trains : List<Train> = new ArrayList<>()

+ TrainsController() {constructor, implicit}
+ getTrain(trainId : String) : Train
+ getTrainInfo(trainId : String) : TrainInfoResponse
+ getStationInfo(stationId : String) : StationInfoResponse
+ getTrackInfo(trackId : String) : TrackInfoResponse
- getStation(stationId : String) : TrainStation
- getTrack(trackId : String) : Track
- getTrack(fromStationId : String, toStationId : String) : Track
+ getAllTrainsInfo() : List<TrainInfoResponse>
- getLoadInfo(loads : List<Load>) : List<LoadInfoResponse>
+ createStation(stationId : String, type : String, x : double, y : double) : void
+ createTrack(trackId : String, fromStationId : String, toStationId : String) : void
+ createTrain(trainId : String, type : String, stationId : String, route : List<String>) : void {throws InvalidRouteException}
+ listTrains(s)() : List<String>
+ listStations(s)() : List<String>
+ listTracks(s)() : List<String>
+ simulate() : void
+ simulateXnumberOfMinutes : int) : void
+ createPassenger(startStationId : String, destStationId : String, passengerId : String) : void
+ createFreight(startStationId : String, destStationId : String, freightId : String, weight : int) : void
+ createRefrigeratedFreight(startStationId : String, destStationId : String, freightId : String, weight : int, minsTillSpoil : int) : void
+ createPassenger(startStationId : String, destStationId : String, passengerId : String, isTechnician : boolean) : void
+ createTrack(trackId : String, fromStationId : String, toStationId : String, isBreakable : boolean) : void
+ hasTrack(a : String, b : String) : boolean
+ isCyclicalRoute : List<String>) : boolean
- isAtStargetStation : TrainStation) : boolean

Train

- FREIGHT_SPEED_REDUCTION_FACTOR : double = 0.0001 {static, readOnly}
- trainId : String {readOnly}
- type : String {readOnly}
- stationId : String
- location : String
- route : List<String> {readOnly}
- position : Position
- maximumLoad : int {readOnly}
- speed : double {readOnly}
- direction : int
- routeIndex : int
- isCyclical : boolean
- currentLoads : List<Load> = new ArrayList<>() {readOnly}
- routeStations : List<TrainStation> = new ArrayList<>()

+ Train(trainId : String, type : String, stationId : String, route : List<String>, x : double, y : double, maximumLoad : int, speed : double) {constructor}
+ getTrainId() : String
+ getType() : String
+ getStationId() : String
+ getLocation() : String
+ getRoute() : List<String>
+ getPosition() : Position
+ getMaximumLoad() : int
+ getSpeed() : double
+ setCyclical(isCyclical : boolean) : void
+ isCyclical() : boolean
+ getTargetStationId() : String
+ getCurrentLoads() : List<Load>
+ getCurrentLoadWeight() : int
+ setRouteStations(routeStations : List<TrainStation>) : void
- getRouteStationPosition(stationId : String) : Position
+ canCarryLoad : Load) : boolean
+ calculateTimeToDestination(destStationId : String) : int
+ canMoveTargetStation : TrainStation, curTrack : Track, isTargetStationFull : boolean) : boolean
+ moveStargetStation : TrainStation, curTrack : Track, isTargetStationFull : boolean) : boolean
onMovedTrack : Track) : void
+ addLoad(load : Load) : void
+ removeLoad(load : Load) : void
+ disEmbarkLoads(stationId : String) : void
+ deteriorateLoads() : void
+ trackDamage() : int
+ repairRate() : int
+ canMoveOnBrokenTrack() : boolean
+ getTechnicianCount() : int
+ compareTo(other : Train) : int {override}
+ carryPassenger() : boolean {abstract}
+ carryFreight() : boolean {abstract}
+ cyclicalRoute() : boolean {abstract}

TrainStation

- stationId : String {readOnly}
- type : String {readOnly}
- position : Position {readOnly}
- maxTrains : int {readOnly}
- currentLoads : List<Load> = new ArrayList<>() {readOnly}

+ TrainStation(stationId : String, type : String, x : double, y : double, maxTrains : int) {constructor}
+ getStationId() : String
+ getStationType() : String
+ getPosition() : Position
+ getMaxTrains() : int
+ getLoads() : List<Load>
+ addLoadOload : Load) : void
+ embarkLoads(train : Train) : void
+ deteriorateLoads() : void

PassengerStation

- MAX_TRAINS : int = 4 {static, readOnly}

+ PassengerStation(stationId : String, type : String, x : double, y : double) {constructor}

FreightStation

- MAX_TRAINS : int = 3 {static, readOnly}

+ FreightStation(stationId : String, type : String, x : double, y : double) {constructor}

CentralStation

- MAX_TRAINS : int = 8 {static, readOnly}

+ CentralStation(stationId : String, type : String, x : double, y : double) {constructor}

SupplyStation

- MAX_TRAINS : int = 6 {static, readOnly}

+ SupplyStation(stationId : String, type : String, x : double, y : double) {constructor}

CommuterTrain

- MAX_LOAD : int = 3500 {static, readOnly}
- SPEED : double = 2.0 {static, readOnly}

+ CommuterTrain(trainId : String, type : String, stationId : String, route : List<String>, x : double, y : double) {constructor}
+ carryPassenger() : boolean {override}
+ carryFreight() : boolean {override}
+ cyclicalRoute() : boolean {override}

FreightTrain

- MAX_LOAD : int = 7000 {static, readOnly}
- SPEED : double = 4.0 {static, readOnly}

+ FreightTrain(trainId : String, type : String, stationId : String, route : List<String>, x : double, y : double) {constructor}
+ carryPassenger() : boolean {override}
+ carryFreight() : boolean {override}
+ cyclicalRoute() : boolean {override}

MaglevTrain

- MAX_LOAD : int = 8000 {static, readOnly}
- SPEED : double = 8.0 {static, readOnly}

+ MaglevTrain(trainId : String, type : String, stationId : String, route : List<String>, x : double, y : double) {constructor}
+ carryPassenger() : boolean {override}
+ carryFreight() : boolean {override}
+ cyclicalRoute() : boolean {override}

RepairTrain

- REPAIR_BONUS_PER_TECH : int = 2 {static, readOnly}

+ RepairTrain(trainId : String, type : String, stationId : String, route : List<String>, x : double, y : double) {constructor}
onMovedTrack : Track) : void {override}
+ trackDamage() : int {override}
+ repairRate() : int {override}
+ canMoveOnBrokenTrack() : boolean {override}

Load

- loadId : String {readOnly}
- destinationStationId : String {readOnly}
- weight : int {readOnly}

+ Load(loadId : String, destinationStationId : String, weight : int) {constructor}
+ getLoadId() : String
+ getLoad() : String
+ getDestinationStationId() : String
+ getWeight() : int
+ getLoadType() : String {abstract}
+ getTechnicianCount() : int
+ getTechnicianCount() : int
+ canBeCarriedBy(train : Train) : boolean {abstract}
+ deteriorate() : void
+ isSpoiled() : boolean
+ canSurvive(ToStraitTime : int) : boolean
+ compareTo(other : Load) : int {override}

Passenger

- PASSENGER_WEIGHT : int = 70 {static, readOnly}

+ Passenger(passengerId : String, destinationStationId : String) {constructor}
+ getLoadType() : String {override}
+ canBeCarriedBy(train : Train) : boolean {override}

Technician

+ Technician(passengerId : String, destinationStationId : String) {constructor}
+ getTechnicianCount() : int {override}
+ getLoadType() : String {override}

Freight

+ Freight(freightId : String, destinationStationId : String, weight : int) {constructor}
+ getLoadType() : String {override}
+ getFreightWeight() : int {override}
+ canBeCarriedBy(train : Train) : boolean {override}

RefrigeratedFreight

- minsTillSpoil : int

+ RefrigeratedFreight(freightId : String, destinationStationId : String, weight : int, minsTillSpoil : int) {constructor}
+ getLoadType() : String {override}
+ deteriorate() : void {override}
+ isSpoiled() : boolean {override}
+ canSurvive(ToStraitTime : int) : boolean {override}

Track

+ MAX_DURABILITY : int = 10 {static, readOnly}
- trackId : String {readOnly}
- fromStationId : String {readOnly}
- toStationId : String {readOnly}
- durability : int
- type : TrackType

+ Track(trackId : String, fromStationId : String, toStationId : String, durability : int, type : TrackType) {constructor}
+ getFromStationId() : String
+ getToStationId() : String
+ getTrackId() : String
+ connects(a : String, b : String) : boolean
+ getDurability() : int
+ getType() : TrackType
+ setType() : boolean
+ setType(type : TrackType) : void
+ decreaseDurOamount : int) : void
+ repairIamount : int) : void
+ updateStatus() : void

onTrack
trains
stations
routeStations
dockedAt (derived)
currentLoads
Tracks
toStation
fromStation
destination
0..*
1
0..1
2..*